Differential Calculus

*Home work sheet # 8

Maxima Minima of function of several variable	***Lagrange Multiplier***
9–20 Locate all relative maxima, relative minima, and saddle points, if any. **9.** $f(x,y) = y^2 + xy + 3y + 2x + 3$ **10.** $f(x,y) = x^2 + xy - 2y - 2x + 1$ **11.** $f(x,y) = x^2 + xy + y^2 - 3x$ **12.** $f(x,y) = xy - x^3 - y^2$ **13.** $f(x,y) = x^2 + y^2 + \frac{2}{xy}$ **14.** $f(x,y) = xe^y$ **15.** $f(x,y) = x^2 + y - e^y$ **16.** $f(x,y) = xy + \frac{2}{x} + \frac{4}{y}$ **17.** $f(x,y) = e^x \sin y$ **18.** $f(x,y) = y \sin x$ **19.** $f(x,y) = e^{-(x^2+y^2+2x)}$ **20.** $f(x,y) = xy + \frac{a^3}{x} + \frac{b^3}{y} \quad (a \neq 0, b \neq 0)$	**5–12** Use Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint. Also, find the points at which these extreme values occur. **5.** $f(x,y) = xy;\ 4x^2 + 8y^2 = 16$ **6.** $f(x,y) = x^2 - y^2;\ x^2 + y^2 = 25$ **7.** $f(x,y) = 4x^3 + y^2;\ 2x^2 + y^2 = 1$ **8.** $f(x,y) = x - 3y - 1;\ x^2 + 3y^2 = 16$ **9.** $f(x,y,z) = 2x + y - 2z;\ x^2 + y^2 + z^2 = 4$ **10.** $f(x,y,z) = 3x + 6y + 2z;\ 2x^2 + 4y^2 + z^2 = 70$ **11.** $f(x,y,z) = xyz;\ x^2 + y^2 + z^2 = 1$ **12.** $f(x,y,z) = x^4 + y^4 + z^4;\ x^2 + y^2 + z^2 = 1$

Taylor Polynomial of Two variable functions

Ex: Determine the 1st and 2nd degree Taylor polynomials $L(x,y)$ and $Q(x,y)$ for the following functions at P

33. $f(x,y) = \frac{1}{\sqrt{x^2+y^2}}$; $P(4,3)$, $Q(3.92, 3.01)$

34. $f(x,y) = x^{0.5}y^{0.3}$; $P(1,1)$, $Q(1.05, 0.97)$

35. $f(x,y) = x \sin y$; $P(0,0)$, $Q(0.003, 0.004)$

36. $f(x,y) = \ln xy$; $P(1,2)$, $Q(1.01, 2.02)$

37. $f(x,y,z) = xyz$; $P(1,2,3)$, $Q(1.001, 2.002, 3.003)$

38. $f(x,y,z) = \frac{x+y}{y+z}$; $P(-1,1,1)$, $Q(-0.99, 0.99, 1.01)$

39. $f(x,y,z) = xe^{yz}$; $P(1,-1,-1)$, $Q(0.99, -1.01, -0.99)$

40. $f(x,y,z) = \ln(x+yz)$; $P(2,1,-1)$, $Q(2.02, 0.97, -1.01)$

**These problems are for the students only as home work. Search the reference books for more examples.*